

ORIGINAL RESEARCH ARTICLE

Single-Cell RNA Sequencing Analysis Reveals a Crucial Role for CTHRC1 (Collagen Triple Helix Repeat Containing 1) Cardiac Fibroblasts After Myocardial Infarction

BACKGROUND: Cardiac fibroblasts (CFs) have a central role in the ventricular remodeling process associated with different types of fibrosis. Recent studies have shown that fibroblasts do not respond homogeneously to heart injury. Because of the limited set of bona fide fibroblast markers, a proper characterization of fibroblast population heterogeneity in response to cardiac damage is lacking. The purpose of this study was to define CF heterogeneity during ventricular remodeling and the underlying mechanisms that regulate CF function.

METHODS: Collagen1 α 1–GFP (green fluorescent protein)–positive CFs were characterized after myocardial infarction (MI) by single-cell and bulk RNA sequencing, assay for transposase-accessible chromatin sequencing, and functional assays. Swine and patient samples were studied using bulk RNA sequencing.

RESULTS: We identified and characterized a unique CF subpopulation that emerges after MI in mice. These activated fibroblasts exhibit a clear profibrotic signature, express high levels of Cthrc1 (collagen triple helix repeat containing 1), and localize into the scar. Noncanonical transforming growth factor- β signaling and different transcription factors including SOX9 are important regulators mediating their response to cardiac injury. Absence of CTHRC1 results in pronounced lethality attributable to ventricular rupture. A population of CFs with a similar transcriptome was identified in a swine model of MI and in heart tissue from patients with MI and dilated cardiomyopathy.

CONCLUSIONS: We report CF heterogeneity and their dynamics during the course of MI and redefine the CFs that respond to cardiac injury and participate in myocardial remodeling. Our study identifies *CTHRC1* as a novel regulator of the healing scar process and a target for future translational studies.

Adrián Ruiz-Villalba,
PhD*

Juan P. Romero, PhD*

Silvia C. Hernández^{1D},
PhD*

:

Volkhard Lindner, MD,
PhD†

David Lara-Astiaso^{1D},
PhD†

Felipe Prósper^{1D}, MD†

*Drs Ruiz-Villalba, Romero, and Hernandez contributed equally and should be considered joint first authors.

†Drs Lindner, Lara-Astiaso, and Prósper contributed equally and should be considered joint senior authors.

The full author list is available on page 1845.

Key Words: fibroblasts ■ myocardial infarction ■ sequence analysis, RNA

Sources of Funding, see page 1845

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Clinical Perspective

What Is New?

- Our study identifies a new subpopulation of cardiac fibroblasts (CFs), reparative CFs, characterized by a distinct transcriptional profile, including *Cthrc1*, with a major role in the fibrotic healing response after myocardial infarction (MI).
- Experiments performed in a mouse model deficient in *Cthrc1* revealed an increase in mortality after MI and a decrease in the fibrotic response.
- We identified CTHRC1⁺ CFs in a swine model of MI and in patients with MI and dilated cardiomyopathy, supporting the conservation of the reparative CF subpopulation in humans.

What Are the Clinical Implications?

- Understanding the role of CFs after MI is highly relevant to identify new therapeutic targets for controlling adverse remodeling after MI.
- The present study provides a unique resource for the cardiovascular field by characterizing the subpopulation of CFs responsible for cardiac repair after cardiac fibrosis and identifying molecular mechanisms implicated in the generation of this subpopulation.
- The conservation of this population in swine and humans and the relation between *Cthrc1* expression and cardiac function after MI suggest that this molecule might be a potential target in patients with MI.

Cardiac fibroblasts (CFs) represent only 10% of the total number of cells in the myocardium; however, they play a critical role in the structural and mechanical maintenance of the heart.^{1,2} After myocardial infarction (MI), CFs become activated, orchestrating a fibrotic response that leads to the generation of a collagen scar, which prevents cardiac rupture.^{3,4} The absence of reliable markers has traditionally hampered our understanding of the specific role of CFs in cardiac homeostasis. Several studies using lineage-tracing reporter strains have allowed us to track the origin and roles of fibroblasts, showing that the fibroblast response to cardiac injury is heterogeneous (reviewed in reference 3). These studies revealed that different CF subtypes may play different roles during the healing process that follows MI.^{5–7} In this context, Fu et al⁸ have recently described a new subpopulation of activated CFs, the matrifibrocytes, which support the mature scar and are characterized by the expression of extracellular matrix (ECM) and tendon genes.

This highlights the need for a better understanding of CF heterogeneity and its effect on processes that mediate repair of the ischemic myocardium. The

development of single-cell RNA sequencing (scRNA-seq) represents an ideal tool to address this knowledge gap. Two recent publications have revealed the presence of 3 subpopulations of CFs in pathologic conditions at single-cell resolution using a sorting and robot-assisted transcriptome sequencing protocol.^{7,9} In accordance with previous studies, both articles show that activated CFs are characterized by the expression of *Postn*. In contrast, Kretzschmar et al¹⁰ described a total of 11 subpopulations of CF after pooling samples from neonatal, adult, and pathologic hearts. All these studies have poor representation of CFs because of the small number of collected cells (63 CFs/185 cardiac cells in reference ⁷, 935 cardiac cells in reference ⁹, and 243 CFs in reference ¹⁰). In contrast, Skelly et al¹¹ performed an unbiased analysis of 10 519 cardiac cells distinguishing 4 subpopulations of CF but only during homeostasis. More recently, Farbehi et al¹² described 8 CF subpopulations within a total of 11 that emerged from unbiased clustering of 16 787 *Pdgfra*-positive cardiac interstitial cells isolated from healthy hearts and hearts 3 and 7 days after infarction. No scRNA-seq study to date has focused exclusively on CFs, their anatomic location, and their dynamics during the course of MI.

We used scRNA-seq to identify a new subpopulation of CFs that mediates ventricular remodeling associated with different cardiac fibrotic processes. This subpopulation is characterized by a specific transcriptomic signature that includes *Cthrc1* (collagen triple helix repeat containing 1), an essential molecule involved in the synthesis and deposition of the fibrotic scar.^{13,14} We describe and characterize the role of some putative regulatory transcription factors (TFs) and validate the noncanonical transforming growth factor (TGF)- β signaling pathway as a mediator of this specific signature. Using a knock-out model for genetic ablation of CTHRC1, we provide evidence for the essential role of this specific fibroblast population during cardiac healing. We provide evidence of the conservation of this CTHRC1⁺ CF subpopulation in a preclinical swine model of myocardial ischemia and in patients with different types of cardiac fibrosis.

METHODS

All data, analytic methods, and study materials are available from the corresponding author on request.

Animal Models

All animal requisitions, housing, treatments, and procedures were performed according to all state and institutional laws, guidelines, and regulations. All studies were approved by the Ethics Committee for Animal Research at the University of Navarra and the Government of Navarra. Studies conducted at Maine Medical Center Research Institute were approved by its Institutional Animal Care and Use Committee and were compliant with The Guide for the Care and Use of Laboratory Animals.

Human Samples

The study protocol was approved by the Medical Ethics Committee and informed consent was obtained from all patients. For RNA studies, human heart samples were provided by Professor S. Janssens. For histologic analysis, human heart samples were provided by the Maine Medical Center BioBank.

Statistical Analysis

In vitro scratch and morphometric assay experiment statistical significance was analyzed using a nonparametric 1-way analysis of variance with a Kruskal-Wallis post hoc test. For the Kaplan-Meier survival curve, log-rank Mantel-Cox test was used to determine statistical difference between the survival curves of the 2 groups of animals. Quantification of collagen deposition statistical significance was analyzed by an unpaired *t* test.

Statistical analysis to obtain the significance when comparing Z score distributions was done with a 2-sided Wilcoxon rank sum test. Differential expression analysis for normalized values was performed using negative binomial generalized linear models.

A 2-way analysis of variance was used to determine statistical significance between the relative quantitative polymerase chain reaction values in different anatomic regions for *POSTN*, *COL1α1*, and *CTHRC1* in pig samples. For comparison of the echocardiography, resonance, and infarct size between reduced and preserved pigs, a 2-tailed *t* test with a Mann-Whitney post hoc test was used.

Data Resources

All scRNA-seq, bulk RNA sequencing, and ATAC-seq (assay for transposase-accessible chromatin sequencing) data are available in a SuperSeries at the National Center for Biotechnology Information Sequence Read Archive database under accession number GSE132146.

Detailed methods are provided in the [Data Supplement](#), including [Supplemental Methods](#), [Tables I–VI](#), [Figures I–XVI](#), [Movie I](#), and [File](#).

RESULTS

Characterization of CF Heterogeneity

To follow the dynamics of the global CF population during MI, we used *Col1α1*-GFP reporter mice. This strain has been shown to homogeneously label fibroblasts from different origins¹⁵ during homeostasis and in different cardiac fibrosis models.^{1,6,7,16} Histologic analysis of healthy and infarcted myocardium at different time points showed changes in the amount and distribution pattern of GFP (green fluorescent protein)-positive cells around the injured site (Figure 1A). To validate this model, we isolated putative CFs (GFP⁺/CD31⁻/CD45⁻), endothelial cells (GFP⁺/CD31⁺/CD45⁻), and bone marrow-derived cells (GFP⁺/CD31⁻/CD45⁺) from healthy myocardium along with myocardium 7, 14, and 30 days postinfarction (dpi; Figure 1B). No GFP⁺

cells were found in the CD31⁺ or CD45⁺ cell fractions at any time point by fluorescence-activated cell sorting or immunohistochemistry (Figure 1B and [Figure IA in the Data Supplement](#)). No transcriptional markers for CFs were found in these populations of cells. The transcriptional profiles of putative CFs and tail/dermal fibroblasts (GFP⁺/CD31⁻/CD45⁻) were shown to be closely similar ([Figure IB–ID in the Data Supplement](#)). An increased percentage and total number of GFP⁺ cells was observed in the infarct zone (IZ) and border zone (BZ) after MI ([Figure IIB in the Data Supplement](#)). *Col1α1*-GFP⁺/CD31⁻/CD45⁻ cells expressed a set of surface membrane markers and a transcriptomic profile consistent with classic CFs ([Figure IIA and IIC in the Data Supplement](#)) and mEFSK4⁺/CD31⁻/CD45⁻ CFs^{1,17} ([Figure IID and IIE in the Data Supplement](#)). Taken together, these results indicate that cardiac GFP⁺ cells are bona fide fibroblasts, validating the *Col1α1*-GFP model to study CF biology during cardiac repair.

To characterize the fibroblast response after MI at single-cell resolution, we profiled the single-cell transcriptomes of GFP⁺ CFs (Figure 1C). Transcriptomes from each time point (7079 cells in healthy myocardium, 10448 cells at 7 dpi, 8337 cells at 14 dpi, and 6805 cells at 30 dpi) were subjected to quality control filtering and merged into a single data set of 29176 individual cells using a canonical correlation approach ([Figure III in the Data Supplement](#)). Unsupervised clustering of the global dataset revealed 11 clusters of GFP⁺ cells (Figure 1D). Ten of these clusters (A through J) represent different fibroblast subpopulations, characterized by high levels of fibroblast-associated molecules, whereas cluster K comprises a population with high expression of classic pericyte markers¹¹ (Figure 1E and [Figure IVA and IVB in the Data Supplement](#)). Clusters B, D, I, and J showed specific expression profiles, representing potential functional CF subgroups, whereas the remaining clusters (A, C, E, F, G, and H) showed less specific transcriptomic identities, likely reflecting intermediate CF types within a larger population (Figure 1E and [Figure IVC in the Data Supplement](#)). We also compared our sorted GFP⁺ CFs with CFs identified in a recent study at scRNA-seq resolution ([Figure V in the Data Supplement](#)) and found that all GFP⁺ CFs fall within previously identified CFs.¹²

Cluster B Represents a Subpopulation of Activated *Periostin*⁺ Fibroblasts Localized in the Infarcted Myocardium

To explore the potential role of the different clusters during MI, we analyzed changes in the percentage of cells in each cluster. Clusters F and H through K remained constant at the different time points ([Figure VIA in the Data Supplement](#)), whereas cluster B manifested

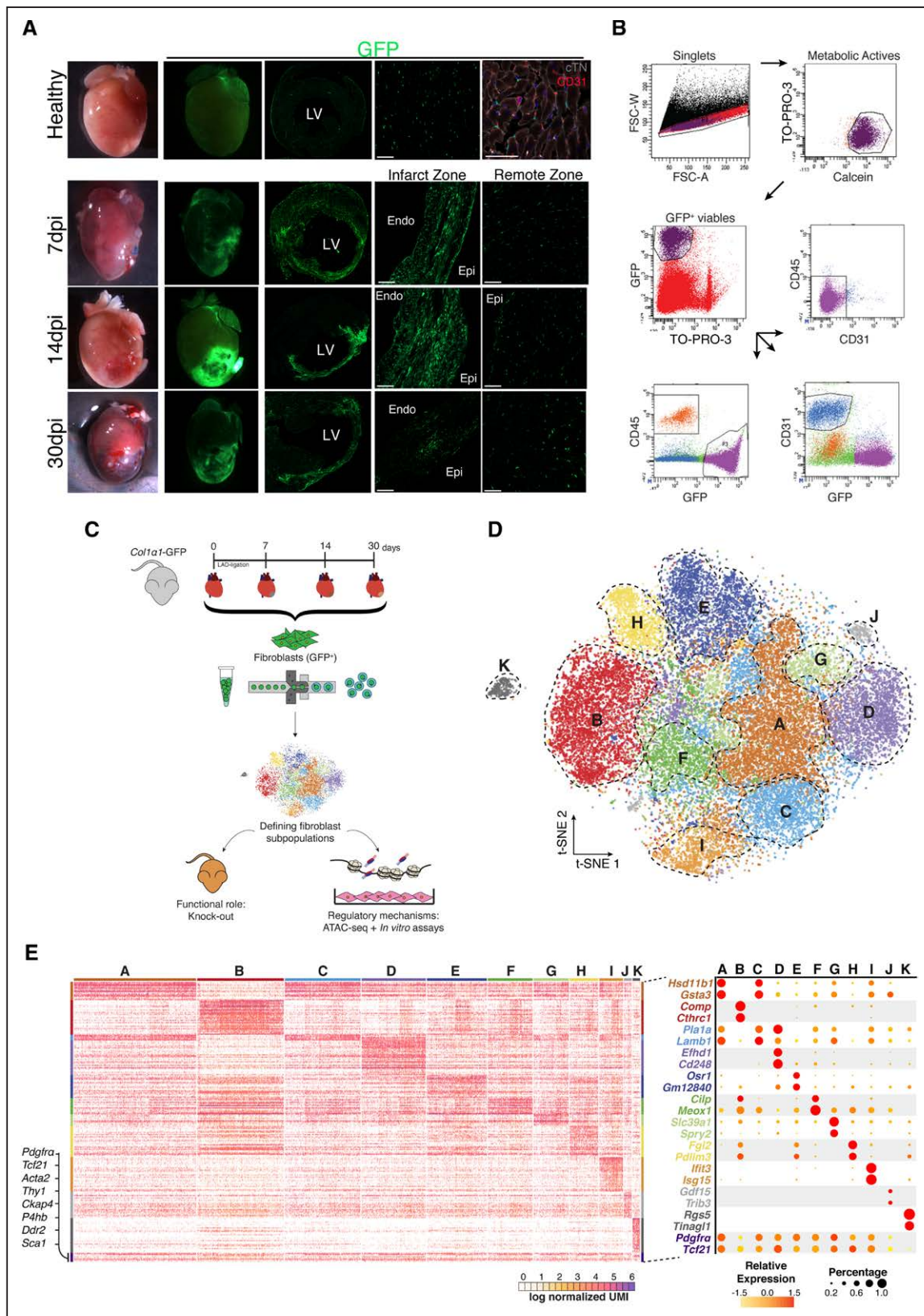


Figure 1. Cardiac fibroblasts (CFs) are a heterogeneous population of cells.

A, Overview of healthy hearts and infarcted hearts at 7, 14, and 30 days postinfarction (dpi) using bright field and fluorescence microscopy; overview and detail of transverse sections show GFP⁺ (green fluorescent protein) cell distribution in the free wall of the left ventricle (LV) in healthy hearts and infarcted hearts 7, 14, and 30 dpi in the infarct and remote zones and detail of GFP⁺ cardiac interstitial cells. Cardiac troponin I⁺ (cTN) cardiomyocytes, gray; CD31⁺ endothelial cells, red; nuclei, blue (DAPI [4',6-diamidino-2-phenylindole]). **B**, Representative gating for isolation of GFP⁺/CD31⁻/CD45⁻ cardiac interstitial cells as performed in healthy hearts and infarcted hearts at 7, 14, and 30 dpi. From left to right: gating for singlets, metabolically active cells (Calcein⁺/TO-PRO-3⁻), (Continued)

Figure 1 Continued. viable cells (GFP⁺/TO-PRO-3⁻), and GFP⁺ cells that are negative for CD45 and CD31. **C**, Schematic representation of the experimental design. *Col1α1-GFP* mice were subjected to myocardial infarction and GFP⁺ CFs were isolated at different time points (healthy hearts and infarcted hearts at 7, 14, and 30 dpi). These cells were used for single-cell RNA sequencing (scRNA-seq). The functional role of CFs was analyzed with a mouse knock-out model and their regulatory mechanisms with assay for transposase-accessible chromatin sequencing (ATAC-seq) and in vitro assays to validate the hypothesis generated. **D**, t-Distributed stochastic neighbor embedding (t-SNE) plot of the global CF population (29 176 cells) comprising the 4 different times. Plot is color-coded by clusters (A through K) identified through unsupervised analysis. Dashed lines delimit clusters. **E**, Heatmap shows log-normalized unique molecular identifiers (UMIs) for scRNA-seq analyzed cells. Top and side bars indicate clusters. Traditional marker genes for CFs are indicated (**left**). Dot plot of expression and specificity of top markers (2) for identified clusters. Dot size represents percentage of cells per cluster expressing the given marker and color represents the relative expression. Scale bars, 100 μm (37 μm in top right picture in **A**). Endo indicates endocardium; Epi, epicardium; and LAD, left anterior descending.

significant changes after MI. Only 2.3% of the total GFP⁺ cells belonged to cluster B in healthy myocardium and the percentage rose sharply after MI (12% at 7 dpi and 34% at 14 dpi) followed by a decrease at 30 dpi (12%; Figure 2A and 2B and Figure VIA in the Data Supplement). Cluster B's transcriptomic signature was significantly enriched in pathways and gene ontology terms related to ECM organization, cell proliferation, and cell-substrate adhesion, in comparison with other clusters (Figure 2C and Figure IVD in the Data Supplement). This was reflected by high and specific expression of structural molecules, including *Fmod* and *Comp*, and by enzymes involved in collagen metabolism including *Ddah1*, *Lox*, and *Ptn*.^{18,19} *Cthrc1*, a gene linked to vascular remodeling and fibrotic processes,^{20–24} was identified as a top marker of cluster B (Figure 2D and Figure VIB in the Data Supplement). Changes in the prorepair expression signature of cluster B fibroblasts were observed at 7, 14, and 30 dpi and were consistent with potential functions involved in specific phases of the repair process (Figure 2E and Figure VIC in the Data Supplement). The highly specific expression of ECM-related genes in cluster B suggests a profound association of this population with the formation of the fibrotic scar.

Immunohistochemical analysis of 3 specific cluster B markers (*Cthrc1*, *Ddah1*, and *Fmod*) revealed that these cells were almost exclusively located in the IZ and BZ at 7, 14, and 30 dpi (Figure 2F and Figure VII in the Data Supplement). In agreement, zonal RNA sequencing profiling of the infarcted heart at 7 dpi showed a clear enrichment in the cluster B signature in the infarct area compared with the remaining clusters (Figure 2G). Taken together, our results identify cluster B as a defined subpopulation of CFs that emerges in response to MI, presents a strong and dynamic profibrotic signature, and localizes to the damaged tissue.

To analyze the origin of cluster B, both RNA velocity and latent time analysis were performed. Using RNA velocity,²⁵ we observed a dynamic movement among all clusters of CFs in healthy heart (Figure VIII in the Data Supplement). At 7 dpi, the previously identified transitions were clearly reduced, and a specific transition from cluster F to cluster B was observed. This transition was prominent at 14 dpi, suggesting that cluster B cells were likely originating from cluster F. At 30 dpi, most of the identified transitions were reduced, including F to B, resembling the dynamics

between subpopulations observed in healthy myocardium (Figure VIIIA in the Data Supplement). Using latent time analysis,²⁶ cluster B cells grouped in regions with the highest latent time values. The expression of *Cthrc1* was almost exclusive to cluster B, whereas other marker genes for quiescent (*Cd90/Thy1* or *Pdgfra/Cd140a*) or activated CFs (periostin [*Postn*] or *Acta2/αSMA*) were expressed in earlier stages of activation (Figure VIIIB in the Data Supplement). This pattern suggests that cluster B fibroblasts correspond to the final activation stage of a subset of *Postn*⁺ CFs generated in response to MI. In this process, expression of *Cthrc1* seems to be specific for the final stages of activation and associated with the healing process through scar formation.

To determine the differences between cluster B cells and other subpopulations of activated CFs,^{4,8} we compared the transcriptomic profile of CFs that express or coexpress the specific markers *Postn* (activated CFs), *Acta2* (matrifibrocytes), or *Cthrc1* (cluster B fibroblasts). Depending on the combined expression of these 3 markers, different subsets of activated CFs appeared (Figure IXA in the Data Supplement). Again, *Cthrc1*⁺ CFs emerged as a subset of *Postn*⁺ CFs, but with enrichment in gene ontology terms related to ECM assembly and organization, as well as collagen fibril organization (Figure IXB in the Data Supplement). In agreement with these results and our scRNA-seq analysis, different subpopulations of activated CFs were detected at 7 dpi by immunohistochemistry (Figure IXC in the Data Supplement). A small number of CTHRC1⁺ CFs were identified at 60 dpi (Figure IXD in the Data Supplement).

To determine whether cluster B can be identified in other models of cardiac fibrosis, we examined the presence of CTHRC1⁺ CFs in a model of chronic cardiac fibrosis generated by infusion of angiotensin II for 28 days.⁵ As shown in Figure X in the Data Supplement, CTHRC1⁺ CFs were detected 2 days after the beginning of treatment with angiotensin II (dpt), with an increase at 7 dpt. At 14 dpt, this population decreased and was no longer detected 28 days after infusion of angiotensin II.

Collectively, these results suggest that cluster B cells represent a specific subpopulation of activated *Postn*⁺ CFs with a major role in reparative cardiac fibrotic processes. Hence, we termed these cells reparative cardiac fibroblasts (RCFs).

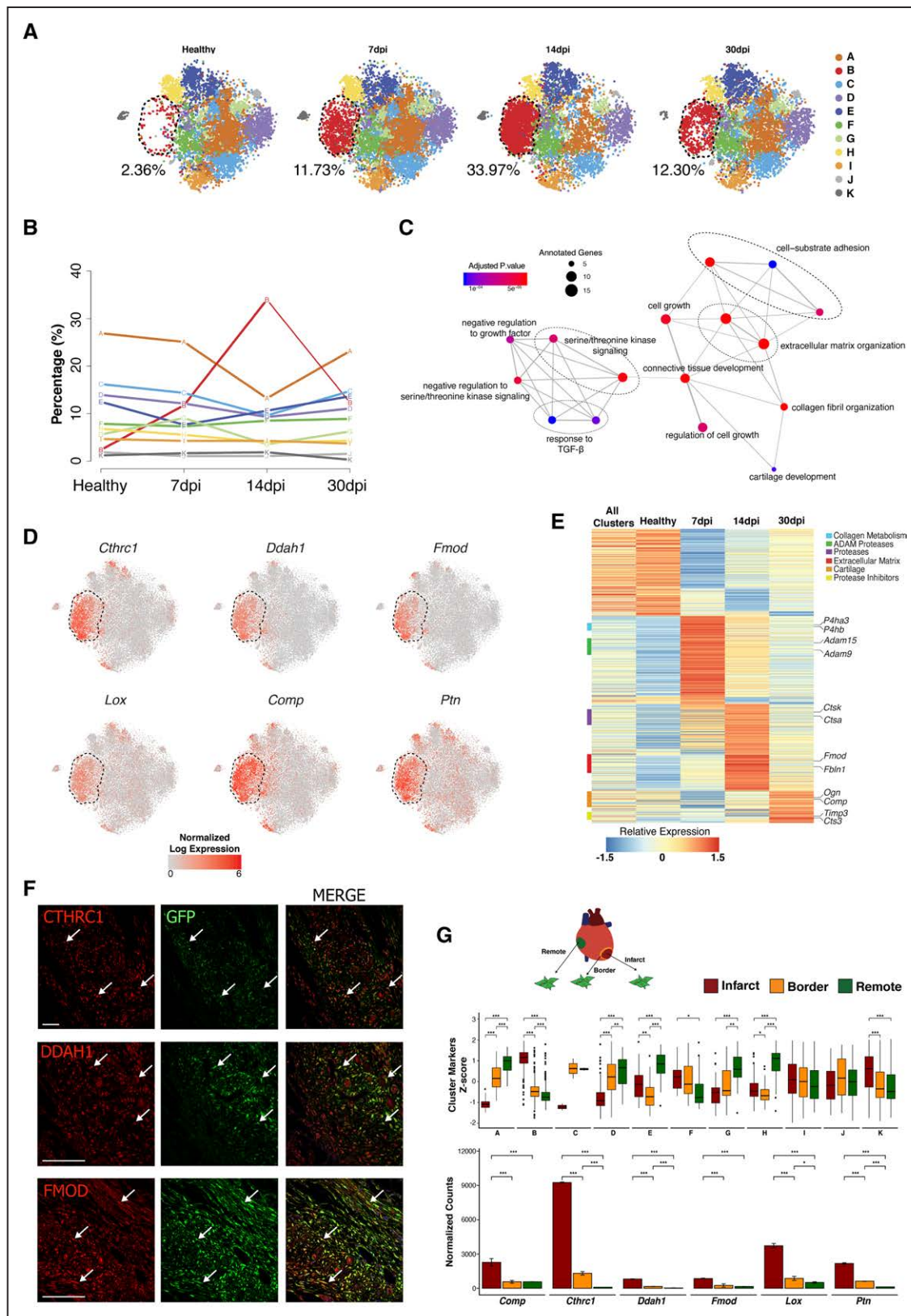


Figure 2. Dynamics of cardiac fibroblast (CF) heterogeneity reveal a unique subpopulation that responds to myocardial infarction.

A, t-Distributed stochastic neighbor embedding (t-SNE) plot shows CF heterogeneity at different time points. Number indicates percentage of cells in cluster B. **B**, Representation shows the dynamics of the percentages of cells by cluster along infarction. Line color-coded by clusters (A through K). **C**, Network representation of enriched pathways based on cluster B markers. Dot sizes represent number of cluster B markers annotated for each pathway and color scale statistical significance for each function. **D**, t-SNE representation of normalized expression of *Cthrc1*, *Ddah1*, *Lox*, *Comp*, *Fmod*, and *Ptn* comprising the 4 time points. Dashed lines delimit cluster B. **E**, Scaled gene expression heatmap shows transcriptional dynamics of cluster B along time points. (Continued)

Figure 2 Continued. **F**, Representative immunohistochemistries of *Col1 α 1*-GFP infarcted hearts at 14 days postinfarction (dpi) show the spatial location of CTHRC1, DDAH1, or FMOD (red) in the infarct zone. GFP⁺ (green fluorescent protein), green; nuclei (DAPI [4',6-diamidino-2-phenylindole]), blue; colocalizations, yellow (arrows). Scale bars, 100 μ m. **G**, Experimental design: CFs (GFP⁺/CD31⁻/CD45⁻) from remote, border, and infarct zones were sorted for bulk RNA sequencing (top). Boxplot shows the Z score distribution of cluster markers in the 3 zones at 7 dpi (Wilcoxon signed rank test, n=2; middle). Bar plot shows normalized expression values for top cluster B markers (bottom; linear model differential expression, n=2). Data are mean $\pm$ SD. Scale bars, 100 μ m. * $P\leq 0.05$, ** $P\leq 0.01$, *** $P\leq 0.001$. ADAM indicates A disintegrin and metalloproteinase; and TGF, transforming growth factor.

Characterization of the Molecular Regulation of the RCF Response

To identify the molecular regulation underpinning the generation of RCFs, we undertook different approaches. First, we leveraged publicly available ChIP-seq (chromatin immunoprecipitation combined with sequencing) datasets to identify TFs whose binding patterns are enriched in the vicinity of RCFs genes. This approach identified several TFs, such as SOX9 and SMAD3 (Figure 3A). Some of the top marker genes of RCFs showed binding motifs for SOX9 (Figure 3B). Sox9 overexpression in cultured CFs induced 23% of the RCFs signatures (28 genes; fold change >1.5, $P<0.05$; Figure 3C), similar to the response observed after incubation of CFs with TGF- β (33 genes; fold change >1.5, $P<0.05$), a classic regulator of CF activation^{27,28} (Figure 3D). These results suggested the potential of SOX9 to promote an RCF phenotype and are consistent with previous studies describing CF activation programs.^{28,29} However, Sox9 overexpression only partially explained the transcriptional signature of RCFs.

To further identify TFs involved in the generation of RCFs, we aimed to profile chromatin accessibility (ATAC-seq) at 7 dpi. We examined the expression patterns of membrane surface markers in our single-cell dataset to identify a specific combination that would allow the isolation of RCFs. None of the classic markers of fibroblasts such as *Thy1* (*Cd90*), *Pdgfra* (*Cd140a*), or *Pgip1* (*Cd44*)^{17,30} was specific for RCFs (Figure 3E). Only *Ox-2/Cd200* was expressed in 62.8% of cluster B cells and in 50% of cluster K cells, which comprise RCFs and pericytes, respectively. The percentages of *Cd200* were substantially lower in other clusters (Figure XIA in the Data Supplement). CD200⁺/GFP⁺ cells colocalized with COL1 α 1 protein in the center of the scar (Figure 3F and Figure XIC in the Data Supplement). We next profiled fluorescence-activated cell sorted CD200⁻ and CD200⁺ fibroblasts at 7 dpi after negative selection for CD146 (*Mcam*) to discard pericytes (Figure 3G). CD200⁺/CD146⁻ CFs showed a significant similarity with the aggregated expression signature of cluster B cells (Figure XIB in the Data Supplement) and significantly higher levels of cluster B, F, G, and H markers in comparison with CD200⁻ CFs (Figure 3H). Using the single-cell data from 7 dpi, 37% of *Cd200*⁺ cells were assigned to RCFs, whereas other clusters showed smaller proportions (Figure XID in the Data Supplement). No combination of surface markers allowed for the unequivocal selection of cluster B CFs, but the expression of *Cd200*⁺/

Cd146⁻ provided the most specific characterization of RCFs. Based on these results, we performed the chromatin accessibility patterns (ATAC-seq) on CD200⁺/CD146⁻ and CD200⁻/CD146⁻ at 7 dpi, as well as on the GFP⁺ CF population at different time points (Figure 4A). The highest chromatin accessibility for RCF-specific loci was observed in CD200⁺/CD146⁻ CFs. Furthermore, a motif enrichment analysis identified several TFs such as RUNX1, SMAD, AP-1(JUN), and TEAD as candidate regulators of RCFs (Figure 4B). With the exception of *Runx1*, most of the single-cell expression patterns of the identified TFs showed no specificity for RCFs (Figure XIIA in the Data Supplement). Although *Runx1* has been reported to play a role in CF activation,³¹ overexpression of *Runx1* in cultured CFs was unable to induce the RCFs-specific transcription signature (Figure 4C). As a result, we considered that the analysis and the characterization of the TF role should be extended to compare the ATAC-seq fingerprint between CD200⁺/CD146⁻ and CD200⁻/CD146⁻ (Figure XIIB in the Data Supplement). Based on motifs detected in peaks from both populations and in the percentage of increased or decreased accessibility, TFs were grouped in 3 different sets: enriched in CD200⁺ peaks but not differentially expressed (set 1); enriched in CD200⁻ peaks but not differentially expressed (set 2); and enriched in CD200⁻ peaks and differentially expressed (set 3). No TFs enriched in CD200⁺ peaks and differentially expressed were found (Figure XIIC in the Data Supplement). To characterize TFs in sets 1 and 3, we used peak-gene mapping to conduct a gene-set analysis per TF. These analyses revealed specific TFs in set 1 (ATF3, JUN, or ZNF93) and in set 3 (RUNX1, WT1, or KLF5; Figure XIID in the Data Supplement). Together, these results represent an initial approach for the characterization of the TFs involved in the generation of RCFs.

As an additional effort to characterize molecular regulation of RCFs, we built a network spanning protein signaling and gene-regulatory interactions using our single-cell expression dataset based on the distances between cluster markers and signaling receptors. This analysis highlighted the noncanonical TGF- β 1/PI3K-Akt pathway over the canonical one as the main driver of RCF gene expression (Figure 4D). To validate this, we stimulated in vitro CFs with TGF- β in the presence or absence of LY294002, a known PI3K-Akt inhibitor.³² This inhibitor abolished the transcriptomic signature of RCFs induced by TGF- β (Figure 4E and Figure XIII A and XIIB in the Data Supplement) as well as the ability of fibroblasts to migrate and proliferate (Figure 4F and the

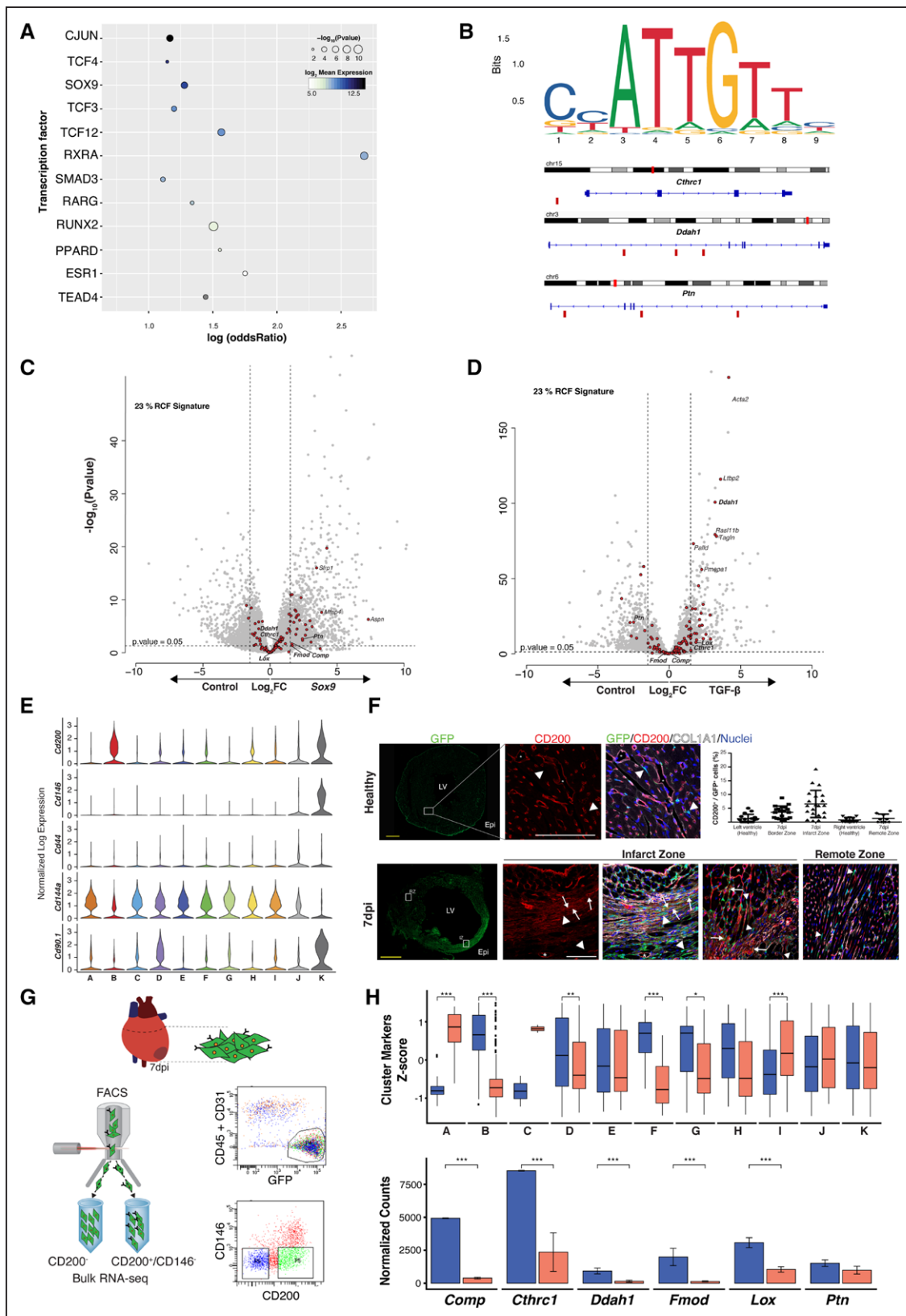


Figure 3. CD200⁺/CD146⁺ cardiac fibroblasts (CFs) provide the most specific characterization of reparative CFs (RCFs).

A, Transcription factor (TF) target gene enrichments. Dot size represents enrichment P value of TF and color represents $\log_2$ transformed expression in CD200⁺ bulk RNA sequencing. **B**, SOX9 DNA binding motif sequence logo and location in some of the top markers for cluster B. Small red rectangles below the gene sequence show confirmed SOX9 binding sites, as determined by JASPAR. **C** and **D**, Volcano plots show differential gene expression of in vitro grown CFs overexpressing Sox9 (left) or treated with transforming growth factor (TGF)- β (right). Genes with $\log_2$ fold change of ± 1.5 and $P < 0.05$ were considered differentially expressed. Red dots represent RCF markers. **E**, Violin plots show single-cell normalized expression of selected surface markers in the pooled CF population (Continued)

Figure 3 Continued. (healthy hearts and infarcted hearts at 7, 14, and 30 days postinfarction [dpi]). **F**, Transverse sections of *Col1α1-GFP* hearts at 7 dpi. Scale bars, 1 mm. Immunofluorescence analysis of GFP⁺ (green fluorescent protein; green), CD200⁺ (red), COL1α1 (gray), and DAPI (4',6-diamidino-2-phenylindole)/nuclei (blue) in healthy left ventricle (LV), and infarct zone (IZ; **left**) and remote zone (**right**) at 7 dpi. Colocalization of GFP⁺ and CD200⁺ in yellow (arrows) in IZ (and colocalized with COL1α1 in light yellow). Arrowheads indicate GFP⁺/CD200[−] cells and asterisks indicate GFP⁺/CD200⁺. Scale bars, 100 μm. Quantification of GFP⁺/CD200⁺ cells in healthy and at 7 dpi (**right, top**). **G**, Gating strategy for isolation of GFP⁺/CD31[−]/CD45[−]/CD200⁺/CD146[−] (CD200⁺) and GFP⁺/CD31[−]/CD45[−]/CD200[−]/CD146[−] (CD200[−]) cardiac interstitial cells at 7 dpi. **H**, Boxplot representation of Z score distributions for cluster markers in CD200⁺ and CD200[−] CF (**top**), Wilcoxon signed rank test, n=2. Normalized expression bar plots (mean±SD) for top RCF markers in both subpopulations (bottom), linear model differential expression, n=2. Data are mean±SD. *P≤0.05, **P≤0.01, ***P≤0.001. FACS indicates fluorescence-activated cell sorting.

Movie I in the Data Supplement). These results emphasize the relevance of the noncanonical TGF-β1/PI3K-Akt pathway in controlling identity and function of RCFs.

To explore the role of TGF-β in depth, we cultured adult CFs in the presence of small molecules that specifically inhibit different elements of the TGF-β1 signaling pathway (Table II in the Data Supplement and Figure XIIC in the Data Supplement). The inhibition of specific elements of the canonical (SMAD2/3)^{33,34} and noncanonical (AKT and p38)^{33,35–38} TGF-β1 signaling pathways reduced the expression of both ECM (*Col1α1*, *Col3α1*, *Lox*) and CF activation (*Cthrc1*, *Ddah1*, *Postn*, *Acta2*)–related genes (Figure XIID in the Data Supplement). However, PD98059, a specific inhibitor of ERK,^{33,34,38} did not affect any of the genes (Figure XIID in the Data Supplement). These results indicate that activation of the profibrotic profile and the phenotypic transformation of RCFs are regulated by a balance between SMAD2/3 and AKT/p38, but not via ERK. Remarkably, SB-431542, a specific inhibitor of TGF-β receptor 1/ALK5,^{33,38,39} selectively reduced the RCF markers together with *Acta2* and *Col1α1*, but not *Postn* and *Col3α1*. This highlights the complexity that underlies the regulation of CF activation.

CTHRC1-Mediated RCF Activity Is Essential for the Formation of the Healing Scar

As *Cthrc1* is exclusively expressed in CFs but not in other cardiac cell populations such as CD45, CD31, or cardiomyocytes (Figure 5A through 5C and Figure XIB and XIC in the Data Supplement), we decided to analyze the functional role of RCFs in mice with genetic ablation of *Cthrc1* (*Cthrc1-KO*; Figure XIVA in the Data Supplement). *Cthrc1-KO* mice appear normal during development and adulthood. No changes were observed in size or weight, and more importantly, no cardiac phenotype was detected.²² However, after MI, knock-out (KO) mice showed a dramatic decrease in survival (80% wild-type [WT] versus 30% KO) attributable to ventricular rupture (Figure 5D). This was associated with a significant decrease in collagen deposition in the free wall of the left ventricle of KO infarcted hearts (50%; Figure 5E). This phenotype was not associated with differences in the transcriptomic profile of other cardiac cell populations between WT and KO mice, suggesting that CTHRC1 had no pleiotropic

effects (Figure XIVC in the Data Supplement). In contrast, we found significant differences between KO and WT CFs in gene ontology terms related to cell division, proliferation, and protein synthesis on MI (Figure 5F and 5G). Cultured KO CFs from healthy mice stimulated with TGF-β1 showed a decrease in gene ontology terms related to angiogenesis, muscle contraction, and vasculature development when compared with WT CFs, indicating a reduced capacity to respond to TGF-β1 (Figure XV in the Data Supplement).

scRNA-seq analysis on *Cthrc1-KO* hearts at 7 dpi (4189 cells) did not show a reduction of RCFs but rather an increased percentage of RCF-like cells in comparison with WT mice (21% versus 14%; Figure 5H and 5I). RCF-like CFs and other subpopulations of activated CFs such as POSTN⁺ were found in the IZ and BZ of *Cthrc1-KO* infarcted mice, indicating that the ventricular rupture phenotype is derived from the absence of CTHRC1 (Figure XIVD and XIVE in the Data Supplement). Accordingly, deletion of *Cthrc1* was associated with downregulation of genes related to ECM organization, collagen biosynthesis, and TGF-β–mediated regulation of the ECM in CFs in comparison with WT mice (Figure XIVF in the Data Supplement). Globally, these results suggested that RCFs orchestrate cardiac repair via secretion of CTHRC1, which affects the deposition and synthesis of ECM molecules and promotes the proliferation of CFs.

An RCF-Like Expression Signature Is Detected in a Preclinical Model of MI and in Human Cardiac Fibrotic Tissue

To assess the translational potential of our findings, we examined whether the RCFs signature can be detected in a preclinical pig model of MI (Figure 6A). This was done by comparing the global transcriptome of representative biopsies from IZ and remote zone of infarcted pig hearts at 8, 60, and 180 dpi (Figure 6B). Despite the lack of cell type resolution, we were able to detect the induction of 33%, 42%, and 32% of the RCF signature at 8, 60, and 180 dpi, respectively, in the IZ but not in the remote zone (Figure XIVA in the Data Supplement). Consistent with murine data, CTHRC1⁺ cells localized to the IZ but not to the remote zone, in the same region where deposits of COL1α1 and POSTN were found (Figure 6C). *CTHRC1* expression was lower at 180 dpi in comparison with 8 and 60 dpi (Figure 6B) with no CTHRC1⁺ cells detected at 180 dpi by immunohistochemistry (data not

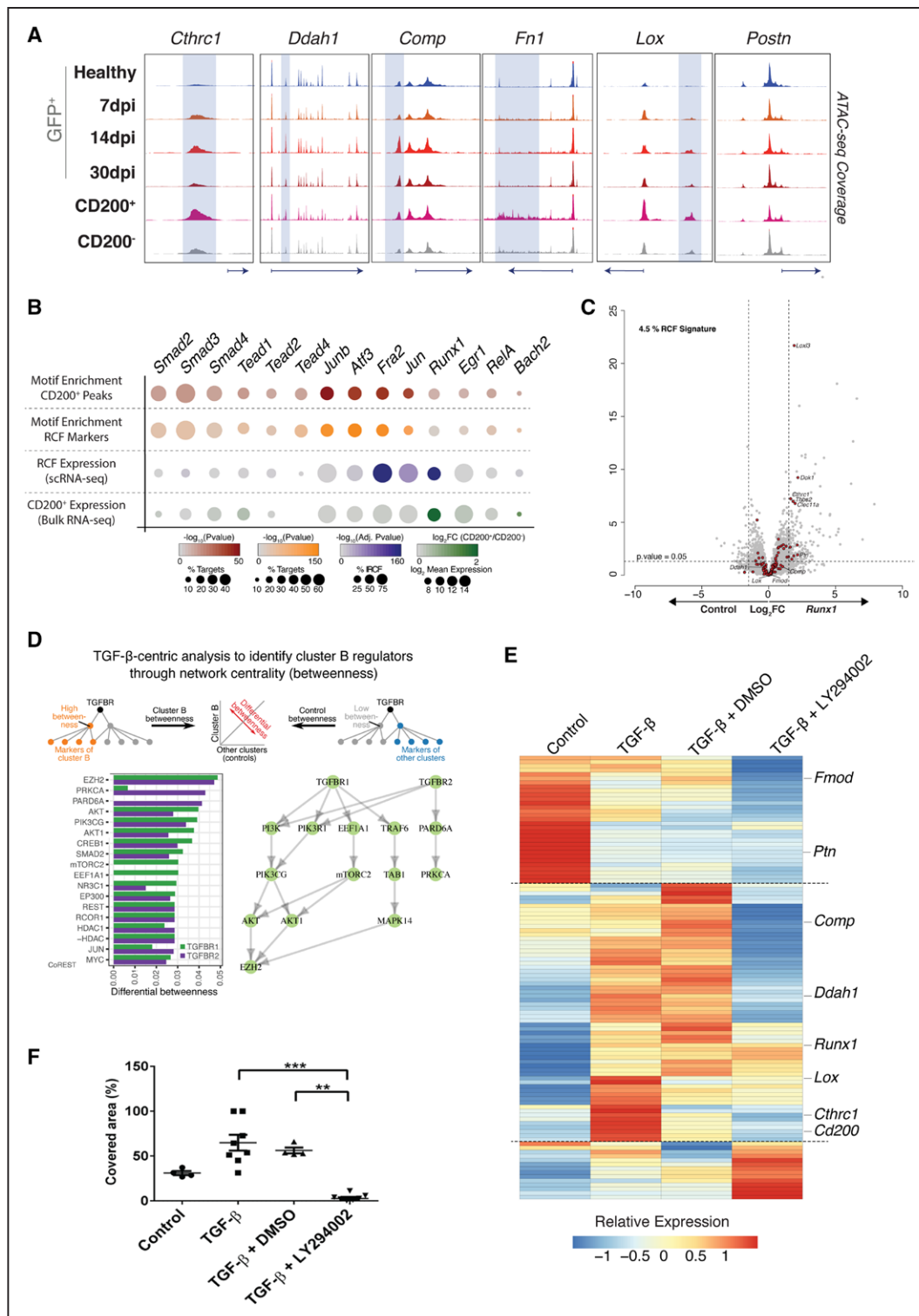


Figure 4. Molecular regulation of reparative cardiac fibroblasts (RCFs) identity.

A, Genome browser snapshots show accessibility profiles of representative loci in the global cardiac fibroblast (CF) population at different time points and in CD200⁺ and CD200⁻ CF subpopulations at 7 days postinfarction (dpi). Shaded areas mark distal regulatory elements displaying increased accessibility in the CD200⁺ subpopulation. **B**, Dot plot represents motif enrichment and expression specificity of potential transcription factor (TF)-mediating RCF response (cluster B/CD200⁺ CF). First row, analysis of CD200⁺-specific accessible distal regulatory elements (± 1.5 Kb from the transcriptional start site); second row, analysis of distal regulatory elements found within RCF-specific loci; third row, expression values of TF in single-cell RNA sequencing (scRNA-seq); fourth row, expression values for TF in CD200⁺ bulk RNA sequencing. **C**, Volcano plot shows differential gene expression of in vitro grown CF overexpressing *Runx1*. Genes with log fold change of ± 1.5 and $P < 0.05$ were considered differentially expressed. Red dots represent RCF markers. **D**, Transforming growth factor (TGF)- β network centrality analysis revealed noncanonical PI3K-Akt pathway related with RCF markers. **E**, Heatmap shows relative expression of RCF markers in nontreated (control), TGF- β , TGF- β + vehicle (dimethyl sulfoxide [DMSO]), and TGF- β + LY294002 cultured CFs. Linear model differential expression, $n = 4$ per group. **F**, Quantification of area covered by cultured CFs after 23 hours of treatment in wound-healing experiment. One-way analysis of variance, Kruskal-Wallis post hoc test, $n = 4$ to 8. $^{**}P \leq 0.01$, $^{***}P \leq 0.001$. ATAC-seq indicates assay for transposase-accessible chromatin sequencing; and GFP, green fluorescent protein.

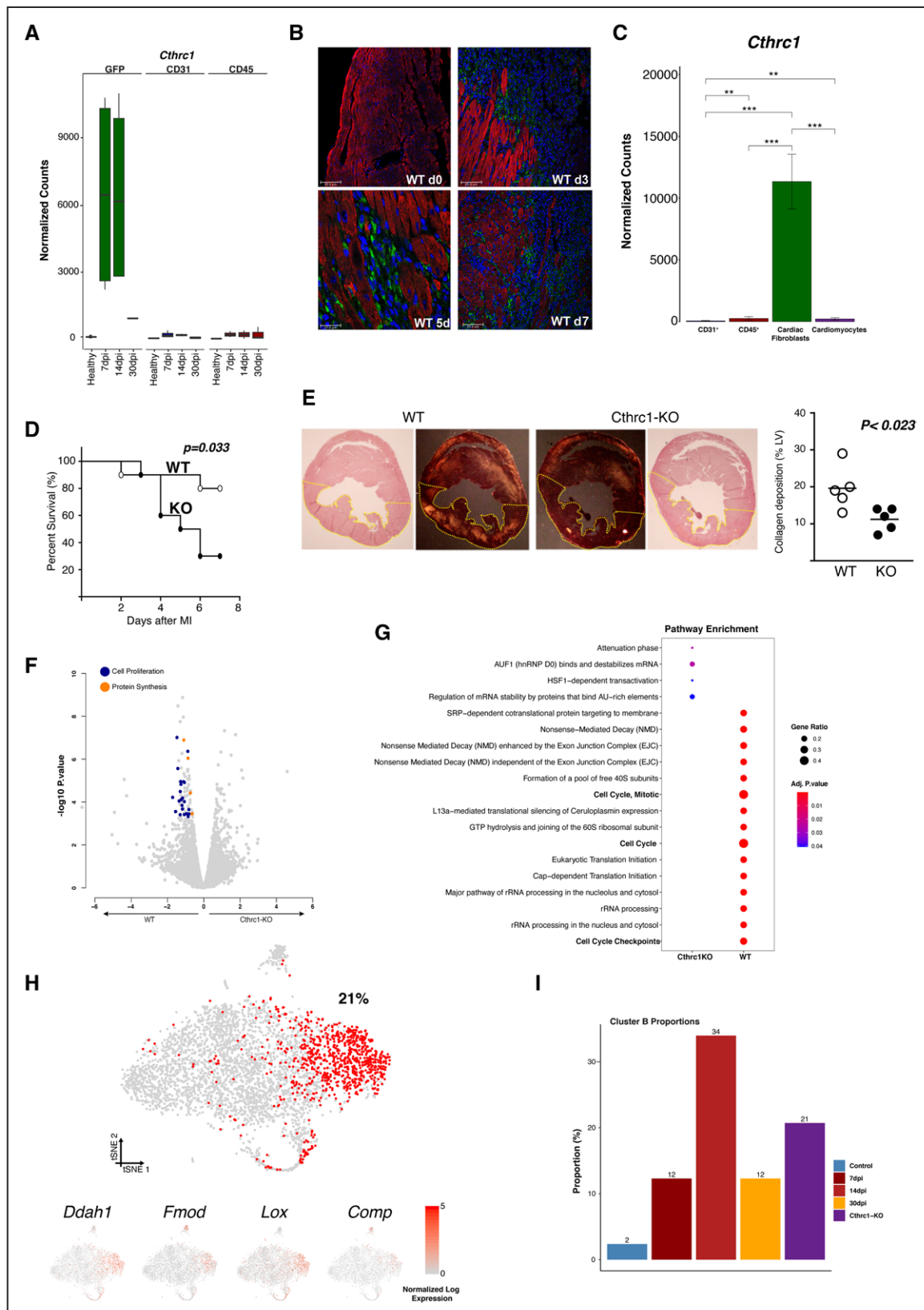


Figure 5. CTHRC1 is an essential effector of reparative cardiac fibroblasts (RCFs) for the healing repair process.

A, Normalized expression bar plot (mean $\pm$ SD) of *Cthrc1* in cardiac fibroblasts (CFs; GFP+/CD31-/CD45-), endothelial cells (CD31+), and bone marrow-derived cells (CD45+) at different time points. **B**, Localization of CTHRC1 (green) in the left ventricle of healthy wild-type (WT) mice and at 3, 5, and 7 days postinfarction (dpi). Cardiac troponin- α cardiomyocytes, red; nuclei (DAPI [4',6-diamidino-2-phenylindole]), blue. **C**, Normalized expression bar plot (mean $\pm$ SD) of *Cthrc1* in endothelial cells (CD31+), bone marrow-derived cells (CD45+), CFs (mEF5K4+/CD31-/CD45-), and cardiomyocytes from WT hearts at 5 dpi; linear model with normalized counts. **D**, Kaplan-Meier survival curves after myocardial infarction (MI) in WT and *Cthrc1* knockout (KO) mice. Log-rank Mantel-Cox test, $n=10$ per group. (Continued)

Figure 5 Continued. **E**, Representative images of collagen deposition in left ventricle of WT (**left**) and KO (**right**) mice. Quantification of collagen deposition in the left ventricle (LV) in both genotypes at 3 dpi (WT, open circles; KO, closed circles), unpaired *t* test (**right**). **F**, Volcano plot shows differential gene expression between WT and KO CFs at 5 dpi. **G**, Dot plot comparison of enriched pathways in the bulk RNA sequencing analysis between *Cthrc1*-KO (left column) and WT (right column) at 7 dpi. **H**, t-Distributed stochastic neighbor embedding (t-SNE) representation of 4189 CFs from 1 KO heart at 7 dpi. Red dots represent RCF-like fibroblasts and the number the percentage of them in relation to the total isolated CF. t-SNE representation for RCF markers (below). **I**, Proportion of cluster B CFs in each of the datasets. ***P*≤0.01, ****P*≤0.001. SRP indicates signal recognition particle.

shown). At 8 dpi, we identified 3 different subpopulations of activated CFs—CTHRC1⁺, αSMA⁺, and αSMA⁺/CTHRC1⁺—in the IZ and BZ, with a decrease at 60 dpi, corroborating the heterogeneity of activated CFs observed in mice (Figure XVIB in the Data Supplement). These results indicated that an RCFs-like population also appears after MI in our preclinical swine model with similar dynamics as in mice.

To assess the relation between the expression of *CTHRC1* and cardiac function, we compared the level of expression of *CTHRC1* in the IZ between animals with midrange, preserved (>45%), and reduced (≤45%) ejection fraction⁴⁰ at 180 dpi (n=12 versus n=9, respectively; Figure 6D). Despite the critical role of *CTHRC1* in the early stages after MI, we found that at later stages (180 dpi) the level of *CTHRC1* (BZ, IZ) was significantly higher in the group with cardiac dysfunction (Figure 6E). Consequently, a negative correlation between the ejection fraction and the expression of *CTHRC1*, *COL1α1*, and *POSTN* in the IZ was found (Figure 6E), which was only statistically significant for *CTHRC1* and *COL1α1*. A positive and significant correlation between the infarct area and expression levels was observed for *CTHRC1* and *POSTN* but not for *COL1α1* (Figure 6F and 6G). Taken together, our results suggest that the upregulation of *CTHRC1* could be beneficial immediately after MI, whereas increased expression at later time points may reflect incomplete repair with progressive fibrosis.

The transcriptomic signature of RCFs was partially found in biopsies obtained from the ischemic zone of patients with ischemic cardiomyopathy and in biopsies from both right and left ventricle of patients with dilated cardiomyopathy. Several genes of the RCF transcriptomic signature (*CTHRC1*, *PTM*, *FMOD*) were significantly overexpressed in all the pathologic conditions included in this study compared with controls, as shown in the bar plots from Figure 6H. Moreover, CTHRC1⁺ CFs were found in the ischemic zone but not in remote zone of infarcted hearts from patients (Figure 6I). These findings highlight the potential role of CTHRC1⁺ RCFs in orchestrating the cardiac repair process in patients with cardiac fibrosis.

DISCUSSION

As the main cellular component responsible for cardiac healing after MI, CFs are an attractive therapeutic target. However, the lack of appropriate markers, the cellular heterogeneity, and the limited understanding of the molecular mechanisms underlying their activation has precluded the development of successful therapies targeting fibrotic

remodeling.⁴ In this study, we characterize the heterogeneity of CFs using *Col1α1*-GFP mice at the single-cell level. We identify a specific subpopulation of CFs defined by the expression of a unique transcriptional signature responsible for the healing response after cardiac injury. Among the top markers, we identified *Cthrc1*, a crucial molecule involved in the ventricular remodeling process. In a recent study, a similar role for *Cthrc1*⁺ fibroblasts has been described in fibrotic lungs.⁴¹ We also identified RCFs in a large swine model and in patients with cardiac diseases that lead to myocardial fibrosis.

Nonmyocyte cardiac cell heterogeneity has been recently characterized with a particular focus on CFs.^{7–10,12} Most of these studies focus on the early phase after cardiac injury¹² or assess other pathologic conditions.^{7,9,10} In the study by Kretzschmar and colleagues,¹⁰ the transcriptome of 282 CFs at 14 dpi was studied, identifying 11 clusters, 1 of which was defined as activated CFs (*Fstll*⁺). This subpopulation had similar properties as RCFs reported in this study but both subpopulations showed different transcriptomic signatures. Fu et al⁸ recently described a new subpopulation of activated CFs, the matrifibrocytes, characterized by the expression of ECM and tendon genes. Despite expression of several RCF marker genes in the matrifibrocytes, it was not possible to compare both subpopulations at the single-cell level. Moreover, RCFs have a role during the healing scar, in contrast to matrifibrocytes, with their role in supporting the mature scar at later stages. Although markers associated with RCFs have been described in other CF subpopulations,^{7–9,12} our transcriptomic, histologic, and functional results clearly indicate that RCFs are distinct from *Postn*⁺, *Acta2*⁺, or matrifibrocytes. An important difference between our work and previous studies is that we focused exclusively on the CF population, unlike recently reported scRNA-seq studies in models of MI.^{9,10,12} This distinction together with the careful description of the anatomic location and the temporal dynamics of RCFs after MI make for a better characterization of CFs heterogeneity and a refined definition of new subpopulations.

Understanding the regulation of RCFs will be essential for the identification of therapeutic strategies. On the one hand, our transcriptomic studies together with our binding site analyses indicate a potential role for different TFs in the generation of RCFs. We validated the role of SOX9, consistent with recent publications.^{28,42} Moreover, our ATAC-seq data revealed TFs that could be involved in the regulation of RCF gene markers and identify other TFs involved in the regulation of CD200⁺ cells through chromatin accessibility.

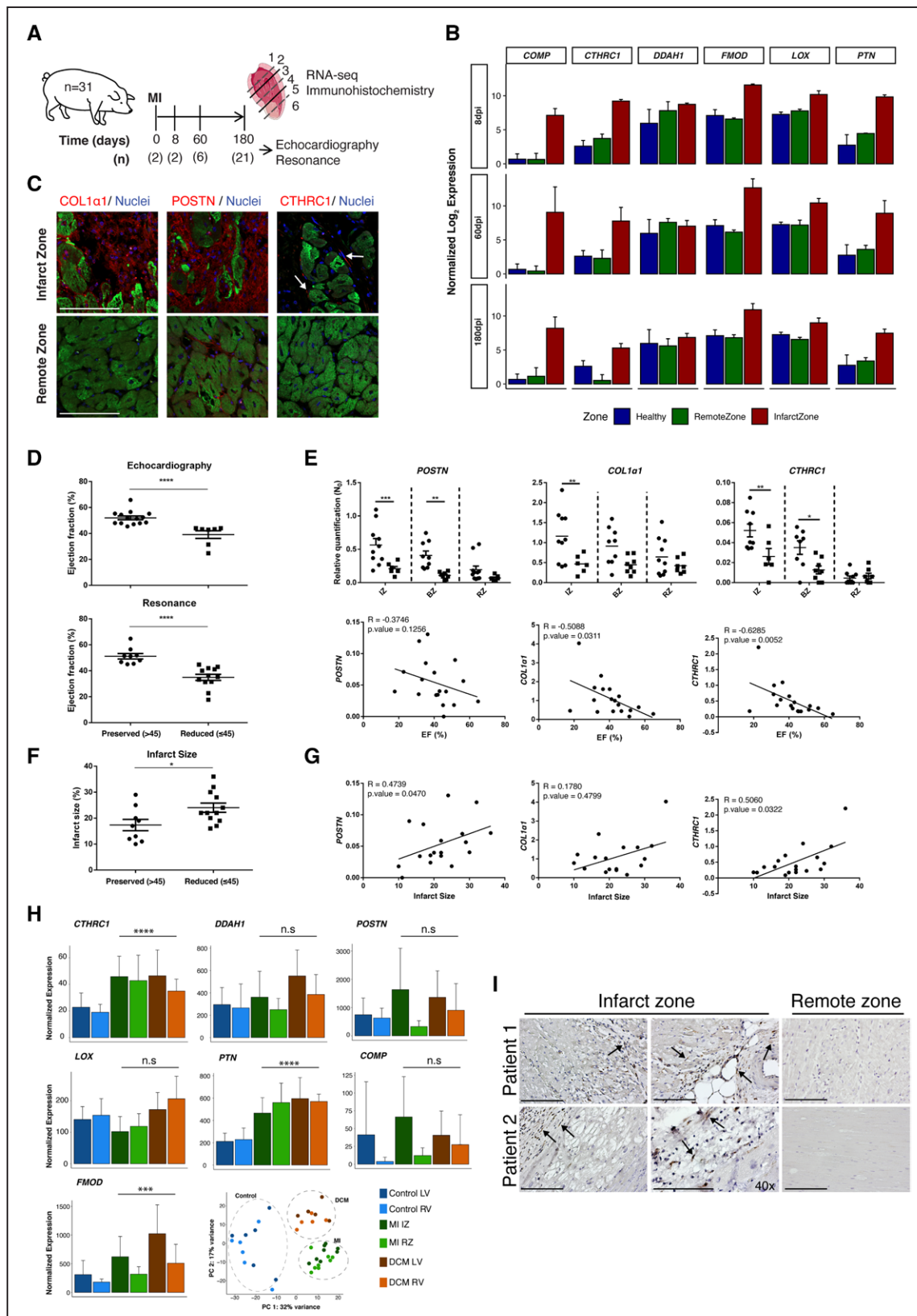


Figure 6. Reporative cardiac fibroblasts (RCFs) marker expression correlates with cardiac function in infarcted pigs and is conserved in humans.

A, Twenty-nine pigs underwent ischemia/reperfusion surgery and 2 animals were used as controls. Cardiac function was determined in 21 pigs after 6 months using echocardiography and MRI. **B**, Normalized expression bar plots (mean±SD) for zonal transcriptomic profiling at 8 days postinfarction (dpi; top; n=2), 60 dpi (middle; n=6), and 180 dpi (bottom; n=9). Normalized expression bar plots (mean±SD) for top RCF markers for healthy hearts (blue) and hearts at 8, 60, and 180 dpi in remote zone (RZ; green) and infarct zone (IZ; red). **C**, Immunohistochemistry of COL1α1, POSTN, and CTHRC1 in IZ and RZ at 8 dpi. **D**, Distribution of pigs with reduced (≤45%, circles; n=9) or preserved (>45%, squares; n=12) ejection fraction (EF) at 180 dpi by echocardiography or MRI. (Continued)

Figure 6 Continued. Two-tailed *t* test with a Mann-Whitney post hoc test. **E**, Comparison of *POSTN*, *COL1a1*, and *CTHRC1* expression levels in different anatomic regions between infarcted pigs with reduced or preserved EF. Two-way analysis of variance, *n*=8 to 10 (**top**). Correlation between EF and expression level of *POSTN*, *COL1a1*, and *CTHRC1* in IZ. Spearman correlation coefficient (**bottom**). **F**, Distribution of pigs with reduced ($\leq 45\%$, circles) or preserved ($>45\%$, squares) infarct area at 180 dpi by magnetic resonance imaging. Two-tailed *t* test with a Mann-Whitney post hoc test, *n*=21. **G**, Correlations between infarct area and expression level of *POSTN*, *COL1a1*, and *CTHRC1* in IZ. Spearman correlation coefficient. **H**, Normalized expression bar plots for top RCF markers (*CTHRC1*, *DDAH1*, *POSTN*, *FMOD*, *LOX*, *PTN*, and *COMP*) in human samples from healthy hearts (left ventricle [LV] and right ventricle [RV]; *n*=6), infarcted hearts (IZ and RZ; *n*=8), and hearts with dilated cardiomyopathy (DCM; LV and RV; *n*=5). Likelihood ratio test. Principal component (PC) analysis scatterplot of human samples subjected to transcriptomic profiling. **I**, Representative images of the immunohistochemistry analysis of CTHRC1⁺ CFs (in brown; arrows) performed on sections of the IZ and RZ obtained from the LV of 2 different patients who had myocardial infarction (MI). BZ indicates border zone. **P*≤0.05, ***P*≤0.01, ****P*≤0.001, *****P*≤0.0001. Scale bars, 100 μm.

Our *in silico* approaches emphasize the role of the TGF-β signaling pathway as a master regulator of CF activation and the development of cardiac fibrosis.^{43,44} TGF-β signaling is known to promote myofibroblast formation and ECM production.^{28,29,45–48} Our results demonstrate that both canonical and noncanonical TGF-β signaling pathways have a significant contribution in the activation of RCFs, with a predominant role for the PI3K-Akt pathway in generating RCFs. This pathway upregulates *Cthrc1*, the key marker gene for RCF, and the migratory and proliferative capacities of CFs. These results suggest that the development of specific inhibitors of the PI3K-Akt pathway could define new targets for modulating cardiac fibrosis, although further studies of the downstream mechanisms are required.

CTHRC1 is a secreted protein that participates in collagen matrix synthesis through TGF-β signaling.^{21,22,49} Moreover, the expression of *Cthrc1* has been recently identified as a potential marker for activated CFs in the heart^{7,9,12} and lungs.⁴¹ The longer follow-up in our study and the evaluation of *Cthrc1* deletion in the context of MI may explain why it was not previously identified as a key player in the cardiac repair process. Consistent with other studies where key components of activated CFs were ablated,^{7,10} the deletion of *Cthrc1* was associated with increased mortality attributable to ventricular rupture. In our case, RCFs were still detected in the myocardium of KO mice, together with other subpopulations of activated CFs. Moreover, our *in silico* data indicated that RCFs originate from activated *Postn*⁺ CFs with a specific role in the secretion and deposition of collagen in the healing scar. These results suggest that the ventricular rupture described by others could be an effect of the deletion of CFs that give rise to RCFs.^{36,50,51} Taken together, these data indicate that *Cthrc1* may be an effector of RCFs, promoting the healing process, rather than an inducer of the activation of this population.

Our results in a preclinical model of MI and in myocardial biopsy specimens of patients with ischemic or dilated cardiomyopathy provide additional translational relevance to our findings. Similar to the observations in mice, expression of *CTHRC1* was upregulated in the fibrotic areas with identification of CTHRC1⁺ CFs in these zones and not in the healthy myocardium. Also, the pattern of expression after MI in pigs showed a significant increase in the level of *CTHRC1* in the early stages after MI, followed by a reduction of its expression levels in the

chronic phase. Of note, the presence of more CTHRC1⁺ CFs or higher levels of *CTHRC1* expression, particularly at later time points, was associated with worse cardiac function, which would seem counterintuitive because *CTHRC1* expression is associated with cardiac healing. These findings suggest that both responses depend on ECM deposition by the CTHRC1⁺ CF lineage, such that ECM contributes to acute cardiac repair and prevents rupture, whereas sustained ECM deposition at later stages may reflect pathologic remodeling. Although additional studies are needed, our findings suggest this molecule might be considered as a potential candidate to study “adverse remodeling” versus “myocardial recovery” in heart failure, based on 3 properties of the molecule: the dynamic transcriptome of *CTHRC1*; its essential role on ECM during the early healing response; and its potentially negative effect on cardiac function during later stages of adverse ventricular remodeling.⁵²

Our study, including the largest number of CFs interrogated by scRNA-seq, clearly demonstrates the remarkable heterogeneity of CFs after MI and defines a specific subpopulation, RCFs, characterized by the expression of *Cthrc1*. Insights into molecular regulation and biological function distinctly support the role of *Cthrc1*⁺ CFs in the early healing process after MI. These novel findings should facilitate the search for more personalized treatment through control of the cardiac fibrotic process.

Study Limitations

The current analysis does not provide complete mechanistic insight because regulation based on post-translational modifications of TF was not considered. With the apparent complexity underlying the regulation of CFs, we cannot rule out that other molecules or subpopulations of CF could have an essential role in the ventricular remodeling process. Future studies should address whether deletion of RCFs using Cre-inducible diphtheria toxin receptor transgenic mouse models instead of CTHRC1 null mice results in a different outcome.⁵³

Competency in Medical Knowledge

Unraveling CFs heterogeneity for understanding different cardiac fibrotic processes is critical. In the case of MI, *Cthrc1*⁺ CFs are responsible for a proper healing

response to cardiac injury. Its role in chronic models requires further studies.

Translational Outlook

Further investigations are needed to determine the potential role of CTHRC1 during MI related to cardiac repair and its mechanism of regulation to identify the therapeutic windows that may allow for controlling the size of the fibrotic scar.

ARTICLE INFORMATION

Received October 24, 2019; accepted August 26, 2020.

The Data Supplement is available with this article at <https://www.ahajournals.org/doi/suppl/10.1161/circulationaha.119.044557>.

Authors

Adrián Ruiz-Villalba, PhD; Juan P. Romero, PhD; Silvia C. Hernández¹, PhD; Amaia Vilas-Zornoza, PhD; Nikolaus Fortelny², PhD; Laura Castro-Labrador³, BS; Patxi San Martín-Uriz, PhD; Erika Lorenzo-Vivas⁴, PhD; Paula García-Olloqui, PhD; Marcel Palacio, MD; Juan José Gavira, MD; Gorka Bastarrika, MD; Stefan Janssens, MD, PhD; Ming Wu, MD, PhD; Elena Iglesias, BS; Gloria Abizanda, BVS; Xabier Martínez de Morentin, MSc; Miren Lasaga, MSc; Nuria Planell, PhD; Christoph Bock⁵, PhD; Diego Alignani, PhD; Gema Medal, BS; Igor Prudovsky, PhD; Yong-Ri Jin, PhD; Sergey Ryzhov, MD; Haifeng Yin, PhD; Beatriz Pelacho, PhD; David Gomez-Cabrero, PhD; Volkhard Lindner, MD, PhD; David Lara-Astiaso⁶, PhD; Felipe Prósper⁷, MD

Correspondence

Felipe Prosper, MD, Department of Hematology and Cell Therapy, Clínica Universidad de Navarra, Pamplona, 31008, Spain; or David Lara-Astiaso, PhD, Advanced Genomics Laboratory, Program of Hemato-Oncology, CIMA, University of Navarra, Pamplona, 31008, Spain; or Volkhard Lindner, MD, PhD, Maine Medical Centre Research Institute, Scarborough, ME 04072. Email fprosper@unav.es or david.larastiaso@gmail.com or LINDNV@mmc.org

Affiliations

Program of Regenerative Medicine (A.R.-V., S.C.H., P.G.-O., E.I., G.A., G.M., B.P., F.P.) and Advanced Genomics Laboratory (J.P.R., A.V.-Z., L.C.-L., P.S.M.-U., E.L.-V., D.L.-A.) and Flow Cytometry Unit (D.A.), Program of Hemato-Oncology, Center for Applied Medical Research (CIMA), University of Navarra, Pamplona, Spain. Instituto de Investigación Sanitaria de Navarra (IdiSNA), Pamplona, Spain (A.R.-V., J.P.R., S.C.H., A.V.-Z., L.C.-L., P.S.M.-U., E.L.-V., P.G.-O., E.I., G.A., D.A., B.P., D.L.-A., F.P.). Department of Animal Biology, Institute of Biomedicine of Málaga (IBIMA) Faculty of Science, University of Málaga, Spain (A.R.-V.). Andalusian Center for Nanomedicine and Biotechnology (BIONAND), Campanillas, Málaga, Spain (A.R.-V.). CeMM Research Center for Molecular Medicine of the Austrian Academy of Sciences, Vienna, Austria (N.F., C.B.). Departments of Cardiology (M.P., J.J.G.), Radiology (G.B.), and Hematology and Cell Therapy (B.P., F.P.), Clínica Universidad de Navarra, Pamplona, Spain. Department of Cardiovascular Sciences, Clinical Cardiology, KU Leuven, Belgium (S.J., M.W.). Translational Bioinformatics Unit (TransBio), NavarraBiomed, Pamplona, Spain (X.M.d.M., M.L., N.P., D.G.-C.). Department of Laboratory Medicine, Medical University of Vienna, Austria (C.B.). Centro de Investigación Biomédica en Red de Cáncer (CIBERONC), Madrid, Spain (D.A.). Maine Medical Center Research Institute, Scarborough (I.P., Y.-R.J., S.R., H.Y., V.L.).

Acknowledgments

The authors thank D.A. Brenner and T. Kisseleva (University of California San Diego) for the gift of *Col1a1-GFP* mice; S. Savide for her comments in the preparation of this article; and MMC BioBank, a core facility of Maine Medical Center Research Institute.

Sources of Funding

This work was supported by Instituto de Salud Carlos III and Fondo Europeo de Desarrollo Regional funds (PI16/00129, CPII15/00017, PI19/00501), Red de Terapia Celular RD16/0011/0005 and Ministerio de Economía y Empresa (Program RETOS Cardiomesh), ERANET II (Nanoreheart), and the Horizon 2020 Program BRAVE. Dr Ruiz-Villalba is supported by Fondo Social Europeo/Ministerio de Economía, Industria y Competitividad-Agencia Estatal de Investigación/IJCI-2016-30254, and the Spanish Ministerio de Ciencia, Innovación y Universidades (RTI2018-095410-BI00). Dr Fortelny is supported by a fellowship from the European Molecular Biology Organization (EMBO ALTF 241–2017). Dr Bock is supported by a New Frontiers Group award of the Austrian Academy of Sciences and by a European Research Council Starting Grant (EU Horizon 2020 research and innovation program, grant agreement 679146). This work was supported in part by the National Heart, Lung, and Blood Institute of the National Institutes of Health under grants R01 HL136560 and HL146504 (Dr Lindner and Dr Ryzhov). This work also used services of the Physiology Core, Histopathology Core, Confocal Microscopy Core, and Mouse Transgenic Core, which are supported by National Institutes of Health/National Institute of General Medical Sciences grants P30GM106391 and P20GM121301.

Disclosures

None.

Supplemental Materials

Methods

Data Supplement Tables I–VI

Data Supplement Figures I–XVI

Data Supplement Movie I

File

Data Resource: GSE132146

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